

MAJOR PROJECT REGISTRATION FORM

Date:31-07-2026

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	AIM-A2
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: An Optimized Explainable Ensemble Learning Framework for AI-Assisted Early Fetal Health Risk Assessment Using Cardiotocography Data

Abstract

Pregnancy-related complications remain a major concern in prenatal healthcare, making early and accurate fetal health assessment essential for timely clinical intervention and improved maternal and fetal outcomes. Cardiotocography (CTG) is a widely used non-invasive technique for monitoring fetal heart rate and uterine contractions to evaluate fetal well-being during pregnancy and labor. However, manual interpretation of CTG recordings is often subjective, time-consuming, and highly dependent on clinical expertise, resulting in inter-observer variability and the possibility of delayed or inconsistent diagnosis. The primary stakeholders of this project include obstetricians, gynecologists, maternal healthcare professionals, hospitals, and pregnant women who require reliable and timely fetal health assessment.

Existing research has explored various machine learning and deep learning techniques for fetal health classification using CTG data. Although these methods achieve promising predictive performance, many fail to provide transparent explanations for their predictions and are rarely designed as deployable clinical support systems. Moreover, existing solutions generally lack integrated functionalities such as patient management, risk scoring, automated report generation, and clinician-oriented workflows, limiting their adoption in real healthcare environments. Therefore, there is a need for an optimized, explainable, and clinically applicable AI framework that not only improves predictive performance but also enhances transparency, usability, and clinician confidence during fetal health assessment.

This project proposes an **AI-Assisted Clinical Decision Support System (CDSS)** for early fetal health risk assessment using the publicly available UCI Cardiotocography dataset. The proposed framework employs an optimized explainable ensemble learning approach to classify fetal health into **Normal**, **Suspect**, and **Pathological** categories while integrating SHAP-based Explainable Artificial Intelligence (XAI) to provide interpretable prediction explanations. The system further incorporates doctor authentication, patient record management, risk scoring, clinical recommendations, automated PDF report generation, and a hospital-oriented dashboard within a unified web application. The scope of this project is limited to CTG-based fetal health risk assessment using retrospective clinical data and excludes real-time CTG device integration, electronic health record integration, and autonomous clinical diagnosis. The project will be considered successful if it demonstrates improved predictive performance, interpretability, and clinical usability over the selected baseline while achieving strong evaluation metrics, including Accuracy, Precision, Recall, F1-score, ROC-AUC, and Balanced Accuracy, thereby providing reliable AI-assisted decision support for healthcare professionals.

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References (Category specific)

Base Paper:

[1] Early Detection of Fetal Health Conditions Using Machine Learning for Classifying Imbalanced Cardiotocographic Data. **Diagnostics**, 2025.

Supporting Papers:

[2] Ługowski, F., Babińska, J., & Stanirowski, P. J. Application of Artificial Intelligence in Diagnosis and Management of Fetal Growth Disorders: A Comprehensive Review. **Frontiers in Medicine**, 2026.

[3] AI-Driven Advances in Women's Health Diagnostics: Current Applications and Future Directions. **Diagnostics**, 2025.

[4] Ahmed, S. S., & Mahmoud, N. M. Early Detection of Fetal Health Status Based on Cardiotocography Using Artificial Intelligence. **Neural Computing and Applications**, 2025.

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